Blockchain Applications: Practice Exam with Solutions

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Instructions. This practice exam covers Blockchain Analytics (Lectures 1–8): 38 multiple-choice questions, each with four options and one correct answer, one point each, no negative marking. Questions 1–23 test concepts and the reading of course output; questions 24–38 are built on three short cases, each followed by five questions. No calculation beyond one-step mental arithmetic is needed. The real exam allows no books, notes or other materials, lasts 60 minutes and has 30 questions: 15 on Crypto Economics and 15 on Blockchain Analytics, including one case with 5 questions.

All numbers come from the lectures (EOD Historical Data, EODHD, prices to 23 October 2026; Lecture 3 to 30 October 2026). Abbreviations: VaR Value-at-Risk (reported as a positive loss, VaR $= -q$), ES expected shortfall, GARCH generalised autoregressive conditional heteroskedasticity, IGARCH integrated GARCH, HS historical simulation, EWMA exponentially weighted moving average, bp basis point (0.01%), HHI Herfindahl–Hirschman index, AUC area under the receiver operating characteristic curve, CAR cumulative abnormal return, DSR deflated Sharpe ratio, QS quantile score, DM Diebold–Mariano, CRIX CRypto IndeX, MV market value.

1. The estimate $\hat{H} = D \cdot 2^{32}/600$, with D the Bitcoin mining difficulty, measures (a) transactions per second; (b) hashes computed per second to secure the network; (c) the number of active miners; (d) the growth rate of the money supply.

   Answer (b). Difficulty sets how many hashes are needed on average to find a block every 600 seconds; $\hat{H}$ is the implied computational work per second, a measure of the cost of attacking the chain (Lecture 1).

2. The Herfindahl–Hirschman index of the five largest coins is 0.71. This means (a) 71% of coins are stablecoins; (b) the market is close to perfect competition; (c) market value is highly concentrated in a few coins; (d) the five coins have correlation 0.71.

   Answer (c). HHI $= \sum s_i^2$; with equal shares among five coins it would be 0.20. A value of 0.71 means one coin (Bitcoin) dominates.

3. After the 2024 halving, Bitcoin issuance inflation is about 0.82% a year. This follows from (a) a fixed supply rule: the block reward halves every 210 000 blocks, towards 21 million BTC; (b) a decision of the miners each year; (c) the hash rate; (d) the price of Bitcoin.

   Answer (a). Supply follows a known geometric rule written into the protocol; issuance does not respond to demand, price or hash rate (Lecture 1).

4. The Jarque–Bera statistic for daily Bitcoin returns is 29 620. Therefore (a) returns are normally distributed; (b) the variance is infinite; (c) returns are independent; (d) normality is rejected, and a VaR based on the Normal distribution understates crash risk.

   Answer (d). H_0: skewness 0 and kurtosis 3; the statistic is far above the χ^2_2 critical value of 5.99. Heavy tails make Normal quantiles too thin.

5. Ljung–Box $Q(10)$ is 212.7 for squared returns r_t^2 and 20.2 for returns r_t. This shows (a) returns can be predicted profitably; (b) volatility clustering is much stronger than any autocorrelation of returns; (c) returns are independent; (d) returns are normal.

   Answer (b). Large values for r_t^2 mean that large moves follow large moves: volatility is predictable even if the direction of returns is not.

6. Robust variance-ratio tests do not reject weak-form efficiency in any sub-period. The correct conclusion is (a) prices are proven to follow a random walk; (b) there are no bubbles; (c) the data give no evidence of return predictability at the tested horizons; (d) returns are normal.

   Answer (c). Not rejecting H_0 is not proof of H_0. Efficiency in this sense says nothing about bubbles or about the distribution.

7. A log-periodic power law fits the 2017 Bitcoin bubble with $R^2 \approx 0.98$. Hence (a) a good in-sample fit does not give a reliable forecast of the crash date; (b) the crash date is known precisely; (c) the model is rejected; (d) the bubble did not exist.

   Answer (a). Flexible nonlinear curves fit price paths well; a high R^2 in sample says little about how precisely the critical time t_c is estimated, so its forecasting value is limited.

8. On the reselection day of 30 September 2026 the divisor of the CRIX replica is recomputed from 214.67 to 214.83 million, while the index stays at 9 521.15. The divisor is adjusted (a) to convert coin prices into US dollars; (b) to cap the weight of Bitcoin below 50%; (c) so that a change of the basket does not make the index jump: only price changes move it; (d) to remove the past returns of coins that failed.

   Answer (c). $\text{CRIX}_t = \sum_i \text{MV}_{it}/\text{Divisor}_t$; on a rebalancing day the new divisor is the new basket value divided by the current index level, so the index is continuous. Stock indices such as the S&P 500 use the same device (Lecture 3).

9. Five coins bought with equal weights on 1 January 2018 earn a compound annual growth rate (CAGR) of 17.6% to October 2026 if they are the five largest coins of 2026, but 7.1% if they are the five largest coins of 1 January 2018. The gap shows (a) that small coins always beat large coins; (b) that equal weights are better than market-value weights; (c) that the 2018 data are wrong; (d) survivorship bias: choosing coins with information from 2026 overstates what an investor could have earned in 2018.

   Answer (d). The ex post basket uses today's winners (Zcash rose from 9th to 3rd); an index must select constituents only from data known on each reselection date, as CRIX does (Lecture 3).

10. The spot Bitcoin fund IBIT has a tracking error (TE) of 3.5% a year against Bitcoin priced at 16:00 New York time, but 20.6% against Bitcoin's close at midnight, Coordinated Universal Time (UTC). This shows (a) the fund holds futures instead of Bitcoin; (b) TE must be measured against the benchmark at the time the fund is priced; the fund itself did not change; (c) the fund's fee is 20.6% a year; (d) the fund does not follow Bitcoin.

    Answer (b). TE is the annualised standard deviation of the daily return differences. With non-synchronous prices the difference includes hours of Bitcoin trading the fund has not yet seen; the fee shows up in the tracking difference (-0.28% a year), not in TE (Lecture 3).

11. For Bitcoin, GARCH(1,1)-t gives $\hat{\alpha} + \hat{\beta} = 1.000$. This means (a) volatility does not cluster; (b) volatility shocks do not die out in this sample (integrated GARCH); (c) a volatility shock halves in 36 days; (d) returns are independent.

    Answer (b). Persistence at the boundary 1 gives an infinite half-life and no finite long-run variance; the S&P 500 with 0.981 has a half-life of 36 trading days.

12. A 1% VaR model has 45 breaches where 24.5 are expected; the Kupiec test has $p < 0.001$. The model (a) is too conservative; (b) passes the backtest; (c) underestimates risk; (d) has dependent breaches.

    Answer (c). Almost twice as many losses beyond VaR as promised: the VaR is too small. The Kupiec test concerns the frequency, not the clustering of breaches.

13. With Normal-based expected shortfall, realised Bitcoin losses beyond VaR are 26–31% larger than predicted. This means (a) the VaR level is wrong; (b) ES is always larger than VaR; (c) the Normal model is too conservative; (d) the Normal distribution understates how deep losses go once the VaR is breached.

    Answer (d). ES is the mean loss beyond VaR. With heavy tails the losses past the threshold are deeper than a Normal tail implies, even when the breach count is acceptable.

14. During the SVB weekend USDC fell to −1 036 bp, while the value of a token with zero recovery on the SVB deposit was −825 bp. The extra discount reflects (a) a liquidity premium: redemption was impossible while banks were closed; (b) the expected credit loss; (c) a data error; (d) a change of the peg.

    Answer (a). Even the worst credit case explains only −825 bp. The rest is the price of time and uncertainty when arbitrage through redemption is closed.

15. USDC has the smallest 1% quantile of USDT, USDC and DAI (−7.6 bp) but the largest kurtosis (1 045). Together these numbers say (a) USDC is the safest coin in every respect; (b) USDC is usually very close to the peg, but its risk is concentrated in rare, extreme episodes; (c) the data contain an error; (d) USDC deviations are normally distributed.

    Answer (b). A small 1% quantile describes ordinary days; a huge kurtosis means that the few large deviations (SVB) dominate the tail. The risk is in episodes, not in the daily noise.

16. A Curve StableSwap pool with amplification $A = 2000$ quotes USDC at 0.997 while 90% of the pool is USDC. This indicates (a) USDC is safe, since the price is close to \$1; (b) a balanced pool; (c) the pool is becoming one-sided, a sign of stress the price does not yet show; (d) a data error.

    Answer (c). StableSwap keeps the price flat near parity until the pool is almost one-sided; holders dumping USDC into the pool show up in its composition before the price moves.

17. A price-only alarm for stablecoin depegs has AUC 0.90 on test hours but detects none of the 8 test events in time. The explanation is (a) AUC is always misleading; (b) the data are wrong; (c) 8 events are too many; (d) a high AUC comes from hours inside episodes, while an alarm must fire before an episode starts.

    Answer (d). Event-level evaluation (detection before onset, lead time, false alerts per month) measures what a risk manager needs; hour-level AUC does not.

18. In the stablecoin depeg pressure index (DPI) a higher average selected Quantile Lasso penalty $\hat{\lambda}$ indicates (a) stronger co-movement in the lower tail, with fewer but stronger links; (b) more active predictors; (c) lower volatility; (d) a better forecast.

    Answer (a). As in the Financial Risk Meter, a higher selected penalty means stronger tail co-movement; the number of active predictors falls under pressure.

19. The Quantile Lasso equations behind the DPI fit the lower tail of stablecoin deviations well in sample, yet held-out forecast skill is negative. The lesson is (a) the model is wrong; (b) in-sample fit is not out-of-sample forecasting ability; (c) more data always help; (d) the tail is normal.

    Answer (b). In-sample fit describes the past; a forecast must be judged on data not used in estimation.

20. In a Diebold–Yilmaz spillover table, $\mathrm{NET}_{\mathrm{DOGE}} = -18.6$. This means (a) Dogecoin fell 18.6%; (b) Dogecoin causes the other coins; (c) Dogecoin receives more forecast-error variance from the others than it transmits; (d) Dogecoin is uncorrelated with the market.

    Answer (c). NET = TO − FROM; a negative value marks a net receiver. Spillovers are variance shares, not causal effects.

21. The Merge has CAR$[-1,1] = -11.1\%$ ($t = -3.09$); across 13 Ethereum upgrades the mean CAR is -0.18% ($p = 0.92$). The correct conclusion is (a) single upgrades can move the price strongly, but there is no common revaluation; (b) upgrades always destroy value; (c) the Merge result is an error; (d) all upgrades are significant.

    Answer (a). Large reactions of both signs (Merge -11.1%, Pectra $+11.3\%$) cancel in the cross-section.

22. XGBoost classifies tomorrow's Bitcoin direction with 78.3% accuracy in sample and 51.3% out of sample; 50.6% of days are up days. This is (a) strong evidence of predictability; (b) overfitting: the model learnt noise; (c) a data error; (d) proof of market inefficiency.

    Answer (b). Only out-of-sample accuracy counts, and it is less than one point above the base rate of "always up".

23. The best of 810 trading rules has Sharpe ratio 0.84 in 2018–2022; its deflated Sharpe ratio is 0.72. Hence (a) the rule is significant at 1%; (b) the rule will earn 0.84 in the future; (c) the Sharpe ratio is wrong; (d) once the number of trials is counted, the rule is not significant.

    Answer (d). The DSR is the probability that the true Sharpe ratio exceeds the best one expected by chance among 810 tries; 0.72 is below 0.95. After 2022 the rule ranked 779th of 810.

Case 1: Bitcoin Volatility and VaR Backtest

GARCH(1,1)-t	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	half-life	$\hat{\nu}$
Bitcoin	0.144 (0.058)	0.076 (0.012)	0.924 (0.012)	1.000	∞	3.13 (0.15)
S&P 500	0.035 (0.009)	0.162 (0.022)	0.819 (0.021)	0.981	36	5.85 (0.71)

1-day VaR 1%, Bitcoin, 2020–2026	hits	rate (%)	Kupiec p	indep. p	last 250	Basel zone
Normal, 250 days	45	1.83	< 0.001	0.850	5	amber
HS, 500 days	25	1.02	0.924	0.256	3	green
EWMA, Normal	51	2.08	< 0.001	0.952	5	amber
GARCH(1,1)-t	28	1.14	0.491	0.329	2	green

Daily log returns in %; robust standard errors in brackets; 2 453 test days, 24.5 hits expected. Mean VaR: HS 8.64%, GARCH-t 8.12%. Kupiec p: test of a 1% breach rate; indep. p: test that breaches do not cluster.

24. The degrees of freedom $\hat{\nu}$ are 3.13 for Bitcoin and 5.85 for the S&P 500. This shows that (a) Bitcoin volatility is less persistent than S&P 500 volatility; (b) Bitcoin returns have about half the variance of S&P 500 returns; (c) standardised Bitcoin shocks stay heavy-tailed after GARCH filtering; (d) GARCH filtering removes the heavy tails of both return series completely.

    Answer (c). A lower $\hat{\nu}$ means a heavier Student-t tail: Bitcoin shocks remain far from Normal even after volatility clustering is filtered out.

25. Which caveat applies to the Bitcoin persistence estimate $\hat{\alpha}+\hat{\beta} = 1.000$? (a) it can also reflect structural breaks between very different market phases; (b) it proves that Bitcoin volatility never returns to a long-run mean; (c) it is invalid because robust standard errors were used in estimation; (d) it implies a half-life of 36 trading days, exactly as for the S&P 500 index.

    Answer (a). Regime shifts in a sample with very different market phases can push estimated persistence towards 1, so IGARCH is a property of this sample, not a proof about the future.

26. The Normal and EWMA-Normal VaR have Kupiec $p < 0.001$ but independence p of 0.850 and 0.952. The justified reading is (a) their breaches cluster in time, so the volatility filter reacts too slowly; (b) they hold too much capital, since the VaR is breached far too rarely in the sample;

(c) they pass the backtest, since the independence test is not rejected; (d) they breach too often but not in clusters: the Normal quantile is too thin.

Answer (d). About twice the promised 1% breaches with no clustering means the dynamics are fine and the problem is the distribution: Normal quantiles are too thin for $\hat{\nu} \approx 3$.

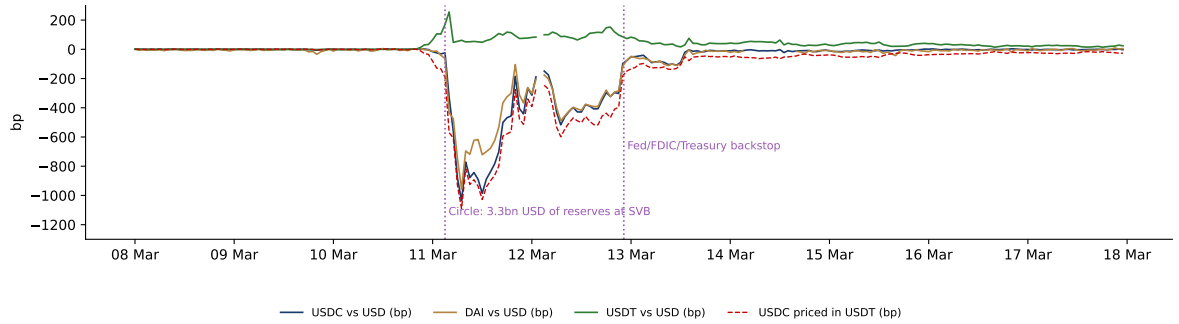
27. A risk manager must choose between HS and GARCH-t, both in the green zone. Which argument favours GARCH-t? (a) its breach rate of 1.14% is closer to the promised 1% than that of HS; (b) it gives correct coverage with a lower mean VaR (8.12% vs 8.64%); (c) it has more breaches in the last 250 days than HS; (d) it is the only model whose Kupiec test is not rejected.

Answer (b). Both models have correct coverage, but GARCH-t reacts to new information daily and needs a lower average VaR, so it ties up less capital.

28. Which further evidence would best separate HS from GARCH-t? (a) a longer list of breach rates computed on exactly the same test days; (b) a Jarque–Bera test of normality on the raw Bitcoin returns; (c) quantile scores with a DM test, plus ES backtests by sub-period; (d) the number of breaches in the last 250 days of the sample.

Answer (c). With 24.5 expected hits, breach-count tests have little power; a consistent loss (QS) with a DM test, ES backtests and sub-period stability compare the models directly.

Case 2: A Stablecoin under Stress



Hourly deviations from \$1 in bp around the Silicon Valley Bank (SVB) failure, March 2023. Circle held 3.3 bn USD of about 40 bn USD reserves at SVB (8.25%). Minimum: USDC −1 036 bp and DAI −955 bp at 07:00 UTC on Saturday 11 March. Early-warning model of the lecture (paper): 72% of test events detected, on average 14 hours ahead, with 2.5 false alerts a month; price-only alarms meet the false-alert budget only by detecting almost no events.

29. If Circle had lost its entire SVB deposit (8.25% of reserves), the fair value of one USDC would deviate from \$1 by (a) −330 bp; (b) −825 bp; (c) −1 036 bp; (d) −211 bp.

Answer (b). With zero recovery each token is backed by $1 - 0.0825 = 0.9175$ USD, a deviation of −825 bp: the worst credit case.

30. USDC traded down to −1 036 bp while about 92% of its reserves were untouched. The episode shows that (a) the remaining reserves were too small to back the token; (b) the market priced exactly the expected credit loss on SVB; (c) a fully backed stablecoin cannot deviate from \$1 for more than a few hours; (d) a peg can fail when redemption is closed, even with most backing intact.

Answer (d). The market went about 211 bp below the worst credit case: over the weekend redemption at Circle was impossible, so the extra discount is a liquidity premium.

31. DAI fell to −955 bp at the same hour, while USDT rose above \$1. The best explanation is (a) much of DAI's collateral was USDC, and holders fled to the other dollar coin; (b) DAI held

its own deposits at SVB, while USDT was backed by the Federal Reserve; (c) DAI and USDT are both issued by Circle, so they moved in opposite ways; (d) the DAI move is a data error, since a decentralised coin cannot depeg.

Answer (a). DAI followed USDC through its balance sheet (contagion via collateral), while USDT gained as a flight to the other dollar stablecoin.

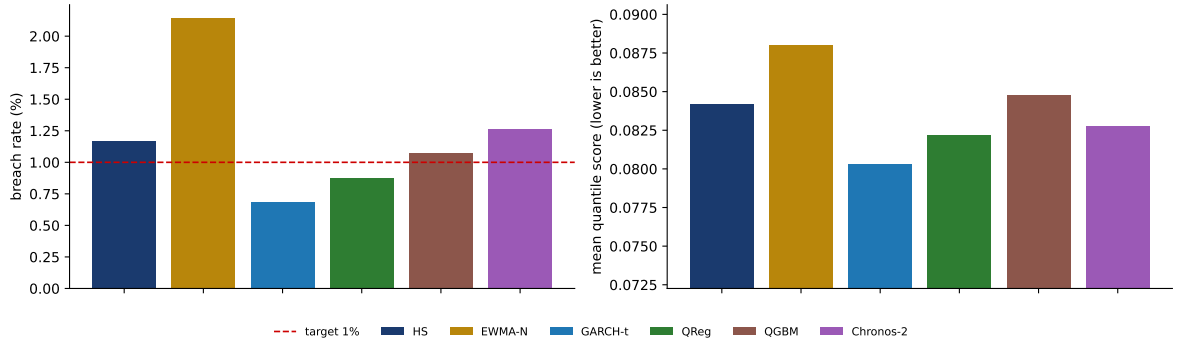
32. The early-warning model detects 72% of test events, on average 14 hours ahead, with 2.5 false alerts a month. The correct reading is (a) 72% of all test hours were classified correctly by the model; (b) each alert is followed by a depeg within the next 14 hours; (c) 72% of test episodes were flagged before they began; (d) the model will detect exactly 72% of future depeg episodes.

Answer (c). The evaluation is at event level (detection before onset, lead time, false alerts per month), not at hour level; with few test events the uncertainty around 72% is large.

33. Why monitor the composition of a StableSwap pool and not only its price, and what is the limitation? (a) the price moves first, but composition is cheaper to observe on chain; (b) composition moves first, but it also shifts for benign reasons (incentives); (c) composition and price move together, so either one is sufficient; (d) composition moves first, and any shift in it signals a coming depeg with certainty.

Answer (b). StableSwap keeps the price near parity until the pool is almost one-sided, so composition leads the price; benign shifts (incentives, rebalancing) mean thresholds must be tested on past episodes.

Case 3: Machine Learning and Risk Forecasts



model	breach %	Kupiec p	QS	DM p*
HS	1.17	0.60	0.0842	0.45
EWMA-N	2.14	0.001	0.0880	0.18
GARCH-t	0.68	0.28	0.0803	–
QGBM (boosted trees)	1.07	0.82	0.0848	0.36
Chronos-2 (pretrained)	1.27	0.41	0.0827	0.54

1-day VaR 1% for Bitcoin, January 2024 – October 2026: 1 027 test days, 10.3 breaches expected. *DM p: test of equal mean QS against GARCH-t. QS: mean quantile score, lower is better. QGBM: quantile gradient boosting machine. Chronos-2 was released in October 2025.

34. GARCH-t has the lowest QS (0.0803) although only 0.68% of days are breaches. This is because (a) a lower QS means more breaches, so GARCH-t is in fact too aggressive; (b) the QS rewards the model whose breach rate is closest to 1%; (c) GARCH-t fails the Kupiec test, so its QS cannot be used; (d) the QS judges daily closeness to the quantile, not the breach count.

Answer (d). The QS is a strictly consistent loss for the quantile: GARCH-t is slightly conservative on average but tracks volatility well day by day.

35. All DM p-values against GARCH-t are 0.18 or higher. The justified conclusion is (a) we do not reject equal mean QS, so the ranking is not conclusive; (b) GARCH-t is significantly better than every other model; (c) all five models give the same forecast on every day; (d) HS is significantly worse than GARCH-t at the 45% level.

    Answer (a). Not rejecting equal predictive accuracy does not prove the models equal, but it means the QS ranking is not supported by significant differences.

36. EWMA-N fails the coverage test (Kupiec $p = 0.001$), yet its QS is not significantly worse than that of GARCH-t (DM $p = 0.18$). Both can hold because (a) the Kupiec test is invalid for models with Normal quantiles; (b) Kupiec counts breaches, while the DM test on QS has little power here; (c) the DM test checks breach clustering, not the breach count; (d) EWMA-N has too few breaches, which the quantile score does not penalise.

    Answer (b). EWMA-N has twice the promised breaches but tracks volatility, so it is close on most days; with about 10 expected breaches in 1 027 days the DM test has little power.

37. Should EWMA-N be used for a 1% VaR? (a) yes, since its QS is not significantly worse than that of the GARCH-t model; (b) no: its volatility filter should be replaced by a constant variance; (c) no: failed coverage is a problem in itself, and the fix is the distribution; (d) yes, as long as the test period is extended to more than 2 000 days.

    Answer (c). Twice the promised breaches means too little capital (amber zone); the remedy is a Student-t or historical quantile, not a different volatility filter.

38. Chronos-2 performs as well as GARCH-t without any fitting to Bitcoin. The main concern about this comparison is (a) its QS of 0.0827 is significantly worse than that of GARCH-t; (b) a pretrained model cannot produce quantile forecasts at all; (c) it was not fitted to Bitcoin, so it cannot contain any look-ahead bias; (d) its training data may include Bitcoin prices from the test period.

    Answer (d). Chronos-2 was released in October 2025, so 2024–2025 prices may be in its training data (look-ahead); only forecasts after the release date are clean.